# Inequalities

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

**Total Questions:** 20 | **Total Marks:** 80 | **Duration:** 1500000 Minutes

## Section: Inequalities

**Problem 1 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

Statements:
D
C, A > B > C, D > E > F,
Conclusions:
I.
E > C
II.
F > B

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (D) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** A > B > C < D > E > F I. Not follow II. Not follow

**Problem 2 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

Statements:
L≤W≤M = J, W = K
Conclusions:
I.
M>K
II.
L = K

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (D) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I: M>K (False) → Given W=K and W≤M, we get K≤M, but M could be equal to K, not necessarily greater. Conclusion II: L=K (False) → Given L≤W and W=K, we get L≤K, but L could be less than K, not necessarily equal.

**Problem 3 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q3 Source: Infosys PYQ Level: Infosys SE Standard

What
should be placed in place of question mark in equation B ≤ A ?N
?K ?S to make A > K definitely true.

A =,
B =,
C <,
D <,
E None

✓ **Verified Answer:** (A) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** B≤A=N>K≥S Since A=N and N>K, we get A>K, which is definitely true. Thus, option (a) is correct.

**Problem 4 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q4 Source: Infosys PYQ Level: Infosys SE Standard

Statement:
3 ≥ 9 < 7 ≤ 10 = 2 ≤ 6
Conclusions:
I.
6 > 9
II.
9 ≤ 2

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (A) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Checking conclusion I: 6 > 9 From the given statement, we get: While moving from 6 towards 9, the common sign of inequalities is '>' and the given conclusions is also '6 > 9'. Clearly, C1 follows. Checking conclusion II: 9 ≤ 2 In the statement 9 < 7 ≤ 10 = 2, the common sign of inequalities between 9 and 2 is '<' whereas the given conclusion is '9 ≤ 2'. Therefore, C2 doesn't follow.

**Problem 5 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

A
@ B means 'A is not less than B'
A
$ B means 'A is not more than B'
A

# B means 'A is neither less nor more than B'

A
* B means 'A is neither more than nor equal to B'
A
% B means 'A is neither less than nor equal to B'
Statements
: R#M$X*T%W@B*
*Conclusions*
*:*
*I.MW*
II.
T@M

A Only

B Only

C Either

D Neither

E Both

✓ **Verified Answer:** (D) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** For conclusion I: We can clearly see the opposite sign persisting between M and W , thus no relationship between them can be established. Hence conclusion I does not follow. For conclusion II: The common sign between M and T is '<', thus M < T or T > M is the actual relationship. Hence conclusion II does not follow

**Problem 6 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q6 Source: Infosys PYQ Level: Infosys SE Standard

Statements:
M>N,
N≥O, O>P
Conclusions:
I.
M>O
II.
N>P

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (E) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Since M>N and N≥O, we can conclude M>O, so Conclusion I is true. Also, since N≥O and O>P, we get N>P, so Conclusion II is true.

**Problem 7 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q7 Source: Infosys PYQ Level: Infosys SE Standard

Statements:
Y
≥ W = N; N ≥ M
Conclusions:
I.
Y > M
II.
Y = M

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (C) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Y ≥ W = N; N ≥ M I. Y > M II. Y = M So either conclusion I or II is follow

**Problem 8 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

Some
friends were discussing about their savings.
M
says that his savings are not more than that of W.
W
says that his savings are neither less than nor equal to S.
S
says that his savings are not less than O, who is not the least
saver.
T
says that his savings are less than M but not less than P.
P
says that his savings are less than M.
Who
among the following is the highest saver?

A W

B S

C O

D P

E None

✓ **Verified Answer:** ✅ **(A) W** **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

**Problem 9 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

Some
friends were discussing about their savings.
M
says that his savings are not more than that of W.
W
says that his savings are neither less than nor equal to S.
S
says that his savings are not less than O, who is not the least
saver.
T
says that his savings are less than M but not less than P.
P
says that his savings are less than M.
P's
savings are less than how many people?

A 1
B 2
C 3
D 4
E More

✓ **Verified Answer:** (E) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

**Problem 10 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

Statement:
M ≥ P < H, V > T = M
Conclusions:
I.
V > P
II.
T ≥ H

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (A) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** M ≥ P < H ..... (i) V > T = M ..... (ii) Combining all statements, we get V > T = M ≥ P < H Thus, V > P is true. Again, T ≥ H is not true.

**Problem 11 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q11 Source: Infosys PYQ Level: Infosys SE Standard

Statements:
F ≤ L > T, N = O ≥ P, F > Q > S
Conclusions:
I.
L > T
II.
F > S

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (E) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I: L>T (True) → Given L>T, this is directly true. Conclusion II: F>S (True) → Given F>Q>S, we get F>S.

**Problem 12 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q12 Source: Infosys PYQ Level: Infosys SE Standard

Statements:
A>B,
B≥C, C>D, DD
II.
C>E

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (A) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** From A>B and B≥C, we get A>C. From C>D, we get A>D, so Conclusion I is true. However, we cannot compare C and E directly, so Conclusion II is false.

**Problem 13 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q13 Source: Infosys PYQ Level: Infosys SE Standard

Some
friends were discussing about their savings.
M
says that his savings are not more than that of W.
W
says that his savings are neither less than nor equal to S.
S
says that his savings are not less than O, who is not the least
saver.
T
says that his savings are less than M but not less than P.
P
says that his savings are less than M.
Which
of the following statements is correct?
I.
O's savings are not more than W.
II.
M's savings are not more than W.
III.
P's savings are not more than W.

A Only

B Only

C Only

D Neither

E All

✓ **Verified Answer:** (B) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** The following equation can be drawn from the given hints. P ≤ T < M ≤ W > S ≥ O

**Problem 14 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q14 Source: Infosys PYQ Level: Infosys SE Standard

'A
© B' means 'A is either smaller than or equal to
B.'
'A
% B' means 'A is neither smaller than nor equal to B.'
'A
@ B' means 'A is either greater than or equal to B.'
'A
$ B' means 'A is neither greater than nor equal to B.'
'A

# B' means 'A is neither greater than nor smaller than

B.'
Statements:
L @ R, R % T, T # M
Conclusions:
I. M $ R II. T $ L

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (E) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I : Conclusion II : Hence, both Conclusions I and II are true.

**Problem 15 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q15 Source: Infosys PYQ Level: Infosys SE Standard

Statements: M = X < Z ≥ W = N ≤ Q < T ≤ V = U
Conclusions:
I. V ≥ W
II. T >/U

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (B) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I: V ≥ W → True because W≤Q

**Problem 16 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q16 Source: Infosys PYQ Level: Infosys SE Standard

**Statement:**
**P**
**S, P < Q, R ≤ Q, R < T**
**Conclusions:**
**I.**
**Q > S**
**II.**
**Q = T**

**A** Only

**B** Only

**C** Either

**D** Neither

**E** Both

✓ **Verified Answer:** (A) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I: Q>S Given P>S and PP>S So, Q>S is true Conclusion II: Q=T Given R≤Q and R

**Problem 17 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

'A
© B' means 'A is either smaller than or equal to
B.'
'A
% B' means 'A is neither smaller than nor equal to B.'
'A
@ B' means 'A is either greater than or equal to B.'
'A
$ B' means 'A is neither greater than nor equal to B.'
'A

# B' means 'A is neither greater than nor smaller than

B.'
Statements:
W $ K, K © G, G @ R
Conclusions:
I.
R $ K
II.
G % W

A Only

B Only

C Either

D Neither

E Both

✓ **Verified Answer:** (B) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusion I: Conclusion II: Hence, only Conclusion II is true.

**Problem 18 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q18 Source: Infosys PYQ Level: Infosys SE Standard

Statement:
P ≤ Q < S = T ≥ U ≥ W < Z
Conclusion:
I.
S > W,
II.
W = T

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (C) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** For conclusion I and II: S > W and W = T From the given statement, we get: S = T ≥ U ≥ W Here, the common sign between S and W is '≥' and the given conclusions are S > W and W = S. Moreover, we are aware that 'S = T' which means we can replace T with S in conclusion 2. Hence, either conclusion I or conclusion II follows.

**Problem 19 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q19 Source: Infosys PYQ Level: Infosys SE Standard

L
≤ O > V = E ≥ S
Which
of the following ones is correct?
1)
L ≤ V 2) O = E 3) O > S 4) S ≥ L

A Only
B Only
C Only
D Only
E None

✓ **Verified Answer:** (C) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** L ≤ O > V = E ≥ S We can compare O and S. which shows that the option 3rd is correct because the common symbol between O and S is '>'.

**Problem 20 • [Marks: +{'positive': 4, 'negative': 1, 'negativeUnattempted': 0}]** ◯ PRACTICE VARIANT

ID: Q20 Source: Infosys PYQ Level: Infosys SE Standard

A
@ B means 'A is not less than B'
A
$ B means 'A is not more than B'
A

# B means 'A is neither less nor more than B'

A
* B means 'A is neither more than nor equal to B'
A
% B means 'A is neither less than nor equal to B'
Statements
: V@I#E*D ; N$E%B#F
Conclusions
:
I.
V%F
II.
V@N

A Only
B Only
C Either
D Neither
E Both

✓ **Verified Answer:** (E) **Step-by-Step Explanation & Derivation:** **Explanation / Solution:** Conclusions: I. V > F II. V ≥ N For conclusion I: From both the statements we get: V ≥ I = E > B = F The common sign between V and F is '>', thus V > F is true. Hence conclusion I follows. For conclusion II: From both the statements we get: V ≥ I = E ≥ N The common sign between V and N is '≥', thus V ≥ N is the actual relationship. Hence conclusion II follows.